Experimental Analysis of Public Policies

Session 8

Synthetic Control, Heterogeneity, and Replication

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Session Objectives

By the end of this session, students should be able to:

- explain how synthetic control constructs a missing counterfactual path;
- choose and audit a donor pool, predictors, weights, and pre-treatment fit;
- interpret post-treatment gaps as causal only under explicit design assumptions;
- use placebo and donor-sensitivity checks as credibility evidence, not automatic proof;
- connect heterogeneity to ATE, ATT, LATE, group-time ATT, RDD, and SCM estimands;
- distinguish credible pre-specified heterogeneity from post-hoc fishing;
- audit replication, robustness, and external validity as part of causal interpretation;
- choose methods by identifying variation rather than by estimator names.

Roadmap

Course Synthesis and Method Choice

Synthetic Control as Constructed Counterfactual

Stress-Testing Synthetic Control

Heterogeneity and External Validity

Replication and Course Close

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From Local Designs to Aggregate Cases

- Session 7 used a threshold to create a local comparison.
- Many major policies affect one city, region, country, or institution.
- There may be no lottery, no sharp cutoff, and few treated units.
- The final question is how to build and challenge a credible counterfactual.

> **Central question**
> How do we synthesise, interpret, and challenge causal evidence?

Method Map: Assignment, Instruments, and Time

Method	Variation	Causal assumption	Target
RCT	Random assignment Z	Assignment independent of potential outcomes	Assignment effect; treatment effect with full compliance
IV	Z-induced receipt D	Independence, exclusion, relevance, monotonicity	LATE for compliers with binary Z, D
DiD	Policy timing and untreated changes	Parallel untreated trends, no anticipation	ATT for treated groups and periods

Choosing a Design

Three scenarios

1. A school lottery allocates scarce tutoring places.
2. A policy is adopted by one region in 2025, with no obvious cutoff.
3. A benefit begins exactly when income falls below a threshold.

Student output: for each scenario, name the likely method, source of identifying variation, key assumption, and local or population estimand.

Three Questions Organise the Final Evidence Audit

1. **What substitutes for the missing counterfactual?**
   Today: a fixed combination of untreated donor units.
2. **What makes that substitution credible?**
   The untreated path must remain predictable after the intervention.
3. **Which effect does the estimate describe?**
   A particular treated unit and horizon, not an automatic population ATE.

> **Then challenge the claim**
>
> Heterogeneity asks where evidence travels.
> Replication checks whether data, implementation and uncertainty support it.

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California: Did a Tobacco-Control Programme Reduce Cigarette Sales?

Proposition 99: approved in 1988; the programme combined a cigarette-tax increase with other tobacco-control measures.

- One treated state; the post-treatment outcome is observed.
- The missing counterfactual is California's sales path **without the programme**.
- California and the rest of the US already had different pre-policy trends.

Two-minute donor-pool discussion

Which states would you exclude if they adopted a similar programme or were affected by California's policy?

Abadie, Diamond and Hainmueller (2010), JASA, pp. 498–499.

The SCM Target and Its Missing Potential Outcome

Let T_0 denote the **first treated period**. For unit 1 at $t \geq T_0$:

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0).$$

Here treatment means the policy path beginning at T_0; state 0 means no policy.

- Observe $Y_{1t} = Y_{1t}(1)$ after adoption; $Y_{1t}(0)$ is missing.
- Untreated donors supply observed $Y_{jt}(0)$, if unaffected by the policy.
- No anticipation makes the treated unit's pre-policy outcomes untreated too.

Population and horizon

This target concerns **one aggregate unit at a particular date**.
It is not an ATE over donor states or a population of individuals.

SCM Workshop: Build the Donor Pool

California: construct the donor pool before interpreting the gaps from 1989 onwards.

1. Which regions are plausible untreated donors?
2. Which regions should be excluded because of spillovers or simultaneous policies?
3. Which pre-treatment predictors should the synthetic unit match?
4. What would make pre-treatment fit unconvincing?

> **Visible output**
> Write a donor-pool rule and one exclusion criterion before seeing any weights.

Donor Pool and Predictors

Choice	Question
Donor pool	Which untreated units could plausibly have received the policy?
Exclusions	Which units face spillovers, anticipation, or simultaneous reforms?
Predictors	Which pre-treatment variables forecast the outcome path?
Pre-period outcomes	Does the synthetic unit track the treated unit before treatment?

Bad-control echo: predictors should be pre-treatment or otherwise unaffected by the policy.

The Equation: A Weighted Untreated California

$$\widehat{Y}_{1t}(0) = \sum_{j=2}^{J+1} \widehat{w}_j Y_{jt}, \qquad \widehat{w}_j \geq 0, \qquad \sum_{j=2}^{J+1} \widehat{w}_j = 1.$$

- Unit 1 receives the policy; units $2, \ldots, J+1$ are untreated donors.
- Choose weights using **pre-treatment** outcomes and predictors.
- Apply those fixed weights to the donors' post-treatment outcomes.

The identifying claim

Absent the programme, the weighted donor path would continue to approximate California's untreated path after treatment.

How Are the Donor Weights Chosen?

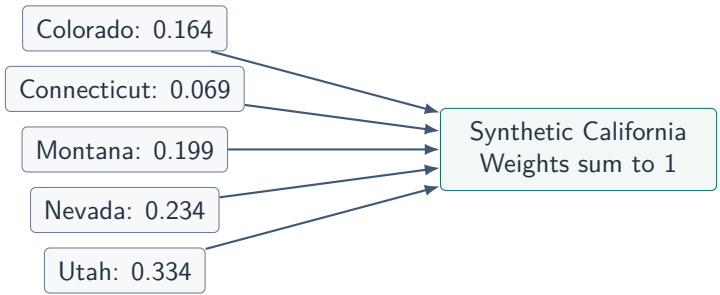
Let P_j collect selected **pre-treatment** outcomes and predictors for unit j.

$$\widehat{w} = \arg \min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \left(P_1 - \sum_j w_j P_j \right)' V \left(P_1 - \sum_j w_j P_j \right).$$

- V is diagonal with non-negative predictor-importance weights, selected using pre-policy information only.
- Non-negative weights summing to one restrict the synthetic predictor vector to the donors' convex hull.
- Inspect what is matched and what is not; a numerical solution need not fit well.

Freeze the design before interpreting post-policy gaps.
Use a pre-policy validation period where feasible to assess prediction beyond the fitting window.

The Published Weights Make the Comparison Visible



The other eligible donor states receive zero weight. These weights construct a comparison, **not an effect**.

Abadie, Diamond and Hainmueller (2010), Table 2. This is a weighting schema, not a DAG.

Convex Hull

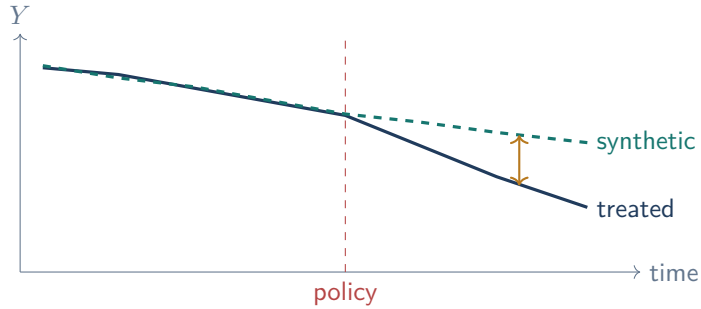
Support problem

Synthetic control cannot construct every possible treated unit. The treated unit must be approximable by weighted untreated units.

- If the treated unit is outside the donor support, fit may be poor.
- Extrapolation is limited by non-negative weights summing to one.
- Poor support weakens credibility even if the post-treatment gap is large.

Connection to Session 2: this is the aggregate-unit version of common support.

Read the Paths: Fit Before the Policy, Gap Afterwards



- Good pre-treatment fit makes the comparison plausible.
- It cannot rule out a new shock affecting only the treated unit afterwards.

Illustrative schema; not estimates from a paper.

The Gap: A Treated-Unit Effect at a Particular Time

$$\widehat{\tau}_{1t} = Y_{1t} - \widehat{Y}_{1t}(0), \qquad t \geq T_0.$$

Published application: by 2000, estimated annual cigarette sales in California were about **26 packs per capita lower** than its synthetic comparison.

Three-minute interpretation

Is this a before–after change, the effect in every US state, or an estimated California counterfactual gap? Would lower recorded sales necessarily mean an identical reduction in consumption?

Abadie, Diamond and Hainmueller (2010), abstract and pp. 500–501.

A Fitted Gap Equals an Effect Plus Counterfactual Error

With no spillovers to donors and $t \geq T_0$:

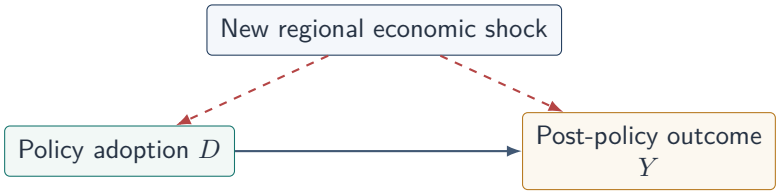
$$\begin{aligned}\widehat{\tau}_{1t} &= Y_{1t} - \sum_j \widehat{w}_j Y_{jt} \\ &= \tau_{1t} + \underbrace{\left[Y_{1t}(0) - \sum_j \widehat{w}_j Y_{jt}(0) \right]}_{\text{unobserved post-policy prediction error}} .\end{aligned}$$

What must carry forward

The donor combination must approximate the treated unit's **untreated** post-policy path. Pre-fit alone does not bound the missing prediction error.

Check: would a new factory closure unique to the treated state appear in this bracket or necessarily in the policy effect?

A New Regional Shock Can Survive Perfect Pre-Fit



Evidence audit: three minutes

A factory closure influences policy adoption and the treated region's outcome after the pre-period. Would matching the earlier path remove this confounding? What information or comparison would help?

Roadmap

Course Synthesis and Method Choice

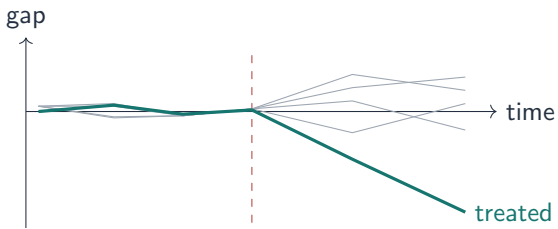
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Placebos: Is the Treated Gap Unusual?



Pretend each donor was treated at the **same policy date**, and build its own synthetic control.
Compare gaps only with attention to **pre-fit quality** and comparable units.

Also remove influential donors and try an earlier placebo date. Each check should address a named threat.

Illustrative schema; not estimates from a paper.

Placebo in Time

- Pretend treatment began earlier than it did.
- A large fake gap before treatment weakens credibility.
- This is analogous to pre-trend and placebo checks in DiD.
- The check probes fit and timing, not all identifying assumptions.

Student output: explain what a large pre-1989 placebo gap would imply for the California example.

Donor Sensitivity

- Drop one influential donor and re-estimate.
- Drop donors exposed to possible spillovers.
- Vary predictor sets and pre-treatment windows.
- Check whether the qualitative conclusion depends on one donor or one modelling choice.

Interpretation

Robustness is convincing when it addresses a named threat, not when it is a long menu of unchecked alternatives.

Inference in SCM

- SCM usually has few treated units, so large-sample formulas are limited.
- Placebo and permutation logic are common.
- Ranks and pseudo-p-values should be interpreted cautiously.
- A donor with poor pre-fit is not a good placebo benchmark.

Reporting habit: show the treated gap, placebo gaps, pre-fit quality, and donor sensitivity together.

SCM vs DiD

	DiD	SCM
Comparison	untreated group trend	weighted donor path
Key assumption	parallel untreated trends	synthetic path approximates $Y_{1t}(0)$
Strength	simple and transparent	visible construction for one treated unit
Threat	different trends	poor donor support or post shocks

Connection: both ask whether untreated units can reveal the treated unit's missing untreated path.

R Mini-Lab: Construct the Path Before Reading the Gap

Question: what counterfactual does the simulated donor pool construct?

```
source("r_exercises/week_08_scm_hte_replication.R")
pre <- wide$year < policy_year
s <- fit_synth(Y_mat, "Y.R1", pre)
gap <- Y_mat[, "Y.R1"] - s$synth
round(s$weights, 3)
c(pre_RMSPE = sqrt(mean(gap[pre]^2)),
  post_mean_gap = mean(gap[!pre]))
```

Five minutes: name the main donors and the post-policy horizon.
Explain why the last number is an estimate, not the identifying assumption.

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Heterogeneous Treatment Effects

- "Does it work?" is rarely the final policy question.
- Effects can differ by baseline need, institution, timing, compliance behaviour, or exposure duration.
- Heterogeneity matters for targeting and scale-up.
- The credibility of a heterogeneity claim inherits the credibility of the underlying design.

> **Policy question**
>
> For whom, where, and under what implementation conditions should the policy be expanded?

Local Estimands Across the Course

Design	Estimand	Locality
Session 1	ATE, ATT, ATU	population or treated status
Session 4	LATE	compliers moved by Z
Session 6	$ATT(g,t)$	cohort and calendar time
Session 7	$\tau_{SRD}(c)$, $\tau_{FRD}(c)$	cutoff and local compliers
Session 8	τ_{1t}	treated aggregate unit and time

External-validity reflex: local credibility is valuable, but it narrows where the estimate travels.

Credible Heterogeneity

- Pre-specify one or two theoretically motivated dimensions.
- Prefer baseline need, pre-treatment predicted exposure, randomised dosage, or eligibility intensity.
- Show subgroup sample sizes and uncertainty.
- Explain why the design remains credible within the subgroup contrast.

Avoid

Conditioning on realised post-treatment exposure can change the estimand or introduce selection bias.

A Subgroup Contrast Is a Causal Target Too

For a baseline characteristic H, define

$$\tau(h) = E[Y_i(1) - Y_i(0) \mid H_i = h].$$

- Comparing $\tau(1)$ with $\tau(0)$ requires identification within both baseline groups.
- In a fully compliant RCT, random assignment and overlap within groups support this interpretation.
- In IV or RDD, subgroup effects can still refer to different complier or cutoff populations.

Before extrapolating: check population, treatment quality, institutions, scale and the distribution of effect modifiers.

Heterogeneity: Put the Difference into the Equation

Let $H_i = 1$ denote high baseline need, measured **before treatment**.

$$Y_i = \alpha + \tau D_i + \theta(D_i \times H_i) + \gamma H_i + u_i.$$

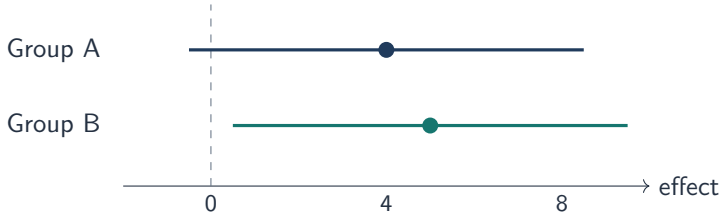
Lower need: treatment contrast τ.
Higher need: treatment contrast $\tau + \theta$.

Difference in contrasts: θ.
Test $H_0 : \theta = 0$, and report its uncertainty.

Causal qualification

The design must identify treatment effects within the relevant baseline groups. Choose a few motivated groups before seeing results; label exploratory searches.

One Significant Estimate and One Insignificant Estimate



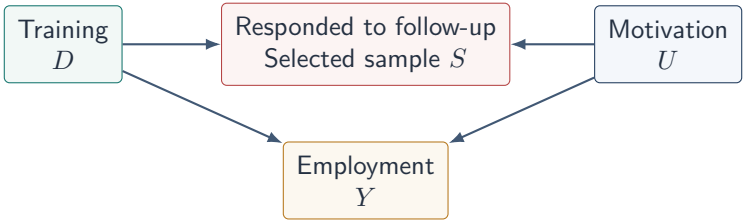
Two-minute vote

One interval includes zero and the other does not. Have we established that the effects differ?

No: estimate the difference and its confidence interval directly. Overlapping intervals alone also do not determine the test of the difference.

Illustrative schema; not estimates from a paper.

Do Not Define Subgroups Using a Post-Treatment Response



Conditioning on survey response can select on a **collider**: $D \to S \leftarrow U$. Treatment and motivation can become associated among respondents. **Prefer:** baseline need or another pre-treatment, substantively motivated dimension.

Multiple Testing

Fishing problem

If many subgroups are searched after the fact, some large differences will appear by chance.

- Limit the number of primary heterogeneity dimensions.
- Report all tested dimensions, not only the most exciting one.
- Treat exploratory heterogeneity as hypothesis-generating.
- Compare subgroup estimates to policy relevance, not only stars.

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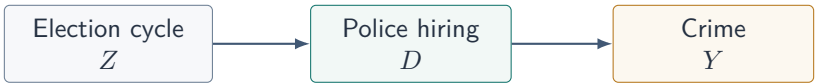
Replication

- Replication asks whether the data, code, specification, and interpretation support the claim.
- It is not only rerunning a regression.
- A replication audit checks design choices and assumptions as well as arithmetic.
- Robustness should be tied to the most credible threat.

> **Final question**
> What evidence would make you trust the result less?

Replication Can Change What the Evidence Supports

Question: do more police reduce crime?



- Levitt's electoral-cycle IV aimed to avoid reverse causality in police hiring.
- McCrary's reanalysis identified implementation and election-dating problems; corrected pooled IV estimates shrank and lost statistical significance.
- Levitt acknowledged errors and presented additional evidence in his reply.

McCrary (2002), AER, 92(4), 1236–1243.

Levitt (2002), AER, 92(4), 1244–1250 (reply).

A Replication Audit Starts with a Specific Threat

Three-minute pair exercise

For the police study or synthetic California, name one check addressing each question:

1. Did code and data implement the claimed comparison?
2. Does the comparison identify the intended counterfactual?
3. Is the estimate precise enough for the policy claim?

A useful robustness check states: the threat, the changed choice, and whether the estimand changes.

Successful rerunning establishes reproducibility of an output. Defending the design requires more evidence.

Replication Audit: Debrief Grid

Audit item	Question
Sample	Who is included or excluded?
Variables	Do Z, D, Y, X, and timing match the claim?
Specification	Is the estimator aligned with the design?
Inference	Are standard errors or placebo checks design-aware?
Robustness	Does each check address a named threat?
Interpretation	Is the estimand local or population-wide?

Specification Curve

- A specification curve displays results across defensible choices.
- It can reveal whether a conclusion depends on one arbitrary specification.
- It should be based on choices justified before looking at the answer.
- It is not a licence to search until a preferred estimate appears.

Connection to robustness: a curve is most informative when each specification reflects a real design uncertainty.

Robustness Checklist

1. What is the main identifying assumption?
2. Which diagnostic most directly probes it?
3. Which alternative specification addresses the main threat?
4. Does the estimand change under the alternative?
5. Does the conclusion survive in magnitude, sign, and policy meaning?

Method Map: Dynamics, Thresholds, and Donors

Method	Variation	Causal assumption	Target
Staggered DiD	Cohort timing	Appropriate untreated trends and clean comparison groups	$ATT(g,t)$, then explicit aggregation
Sharp / fuzzy RDD	Threshold rule	Continuity; fuzzy adds local IV restrictions	Cutoff effect / cutoff-complier LATE
SCM	Treated unit versus fixed weighted donors	Donors reproduce its untreated path; no contaminating shocks/spillovers	Treated-unit effect at time t

Common task: defend the observed comparison as a substitute for the missing counterfactual.

Final Exam Skills

- Translate a policy question into an estimand.
- Identify the missing counterfactual.
- Name the source of identifying variation.
- State the assumption in plain language and notation when useful.
- Interpret the estimate for the correct population.
- Separate identification, estimation, inference, and external validity.

Course Synthesis

The course reflex

1. Where does the identifying variation come from?
2. What assumption makes the comparison causal?
3. What estimand or population does the resulting estimate describe?

- Good empirical work defends a comparison.
- The method name is secondary to the credibility of the counterfactual.
- Local evidence can be powerful when interpreted honestly.

Common Mistakes

Watch for these

- Treating pre-fit, balance, or pre-trends as proof.
- Calling a local estimate an ATE without justification.
- Interpreting subgroup coefficients causally without subgroup identification.
- Treating placebo ranks as generic large-sample p-values.
- Reporting robustness checks that do not address a specific threat.
- Starting with a method name before defining the counterfactual.

Practice Bridge

- Theory sheet: `homework/session_08_optional_homework.md`.
- QCM: `assessments/qcm/session_08_qcm.md`.
- Final exam: `assessments/exams/final_exam_template.md`.
- Mini code task: inspect donor weights, pre-fit, treatment gaps, and placebo gaps.

Student deliverable: a one-page evidence audit: counterfactual, assumption, diagnostic, estimate, inference, external-validity claim, and remaining doubt.

Key Takeaways

- Synthetic control constructs the missing untreated path with weighted donors.
- SCM evidence depends on donor validity, pre-fit, placebo logic, and sensitivity.
- Heterogeneity changes interpretation and external validity.
- Replication is a causal-design audit, not only a coding exercise.
- The best method is the one whose comparison and assumptions you can defend.

Closing line: first defend the comparison; then interpret the estimate.

Appendix: One Model Explaining Why Pre-Fit Can Help

An illustrative untreated-outcome model is

$$Y_{jt}(0) = \delta_t + \lambda_t' \mu_j + \varepsilon_{jt}.$$

λ_t: common time-varying forces; μ_j: unit-specific exposure to them.

$$Y_{1t}(0) - \sum_j w_j Y_{jt}(0) = \lambda_t' \left(\mu_1 - \sum_j w_j \mu_j \right) + \varepsilon_{1t} - \sum_j w_j \varepsilon_{jt}.$$

- If donors reproduce μ_1, common forces cancel at later dates too.
- A long, informative pre-period can help reveal those exposures.
- Noise, weak support or a new treated-specific shock can defeat this argument.

Appendix: Compare Post-Gaps Relative to Pre-Fit

For unit j, let e_{jt} be its actual-minus-synthetic gap. One statistic is

$$R_j = \frac{\sqrt{\text{mean}_{t \geq T_0}\, e_{jt}^2}}{\sqrt{\text{mean}_{t < T_0}\, e_{jt}^2}}.$$

- Refit each placebo using the same rule and a suitable untreated donor pool.
- Declare any pre-fit exclusion rule before comparing post-treatment gaps.
- Near-zero pre-fit error can make a ratio unstable; inspect the paths as well.

Appendix: When Can a Placebo Rank Be a Test?

For M donor placebos, one upper-tail rank statistic is

$$p_{rank} = \frac{1 + \#\{j : R_j \geq R_1\}}{M + 1}.$$

- An exact randomisation interpretation needs exchangeable assignment among the comparison units, a sharp no-effect null and symmetric refitting.
- Donor eligibility and any pre-fit filtering must respect that assignment argument.
- With few placebos, ranks are coarse: the smallest value is $1/(M + 1)$.

Interpretation

Without that assignment argument, report how unusual the treated gap is; the rank is not automatically a randomised-test p-value.

References: Synthetic Control and Replication

- Abadie, A., Diamond, A., and Hainmueller, J. (2010). "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *JASA*, 105(490), 493–505.
  Author-hosted article.
- McCrary, J. (2002). "Using Electoral Cycles in Police Hiring to Estimate the Effect of Police on Crime: Comment." *AER*, 92(4), 1236–1243.
  Author-hosted comment.
- Levitt, S. D. (2002). "Using Electoral Cycles in Police Hiring to Estimate the Effects of Police on Crime: Reply." *AER*, 92(4), 1244–1250.
  Author-hosted reply.